DISCLOSURE DRIFT

Risk disclosures and return forecasting

Research progress report | 10 October 2026

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Primary empirical result	Not estimated
Study status	Pipeline implementation and pilot evidence; empirical completion gates remain open

Research question

Do changes in an issuer's annual Risk Factors disclosures improve forecasts of its next 20-session market-adjusted return beyond momentum, volatility, size and historical industry controls?

This study is inspired by Cohen, Malloy and Nguyen's *Lazy Prices* [1]. It narrows the document input to original annual Item 1A sections and fixes a 20-session forecast comparison. It is a new investigation; the source paper's findings are not results for this sample.

Evidence available now

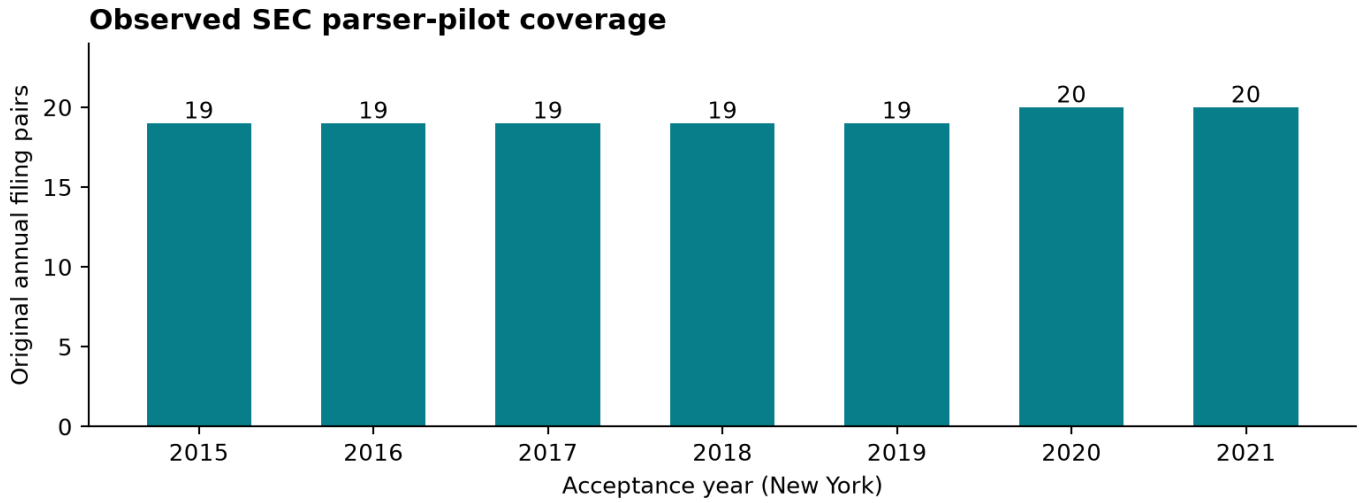
The recovered SEC checkpoint contains **155 original filings, 135 annual pairs and 20 issuers**. The original sources pass the automated audit. Candidate market arithmetic also reproduces, while historical identity, availability, and human reviews remain pending [3-5].

The final 2022-2025 empirical evaluation has not been opened. Figures in this report describe observed SEC text or the frozen method. Any full-pipeline demonstration uses invented data and provides software evidence only.

Public code and audit trail

1. Pilot scope and coverage

The existing sample was selected to exercise document formats, reference exhibits and time conventions. It has no claim to be a representative investment universe. Pair coverage is shown by acceptance year below. The pairing rule preserves issuer identity and consecutive annual filings; predecessor issuers are not silently spliced into the record.



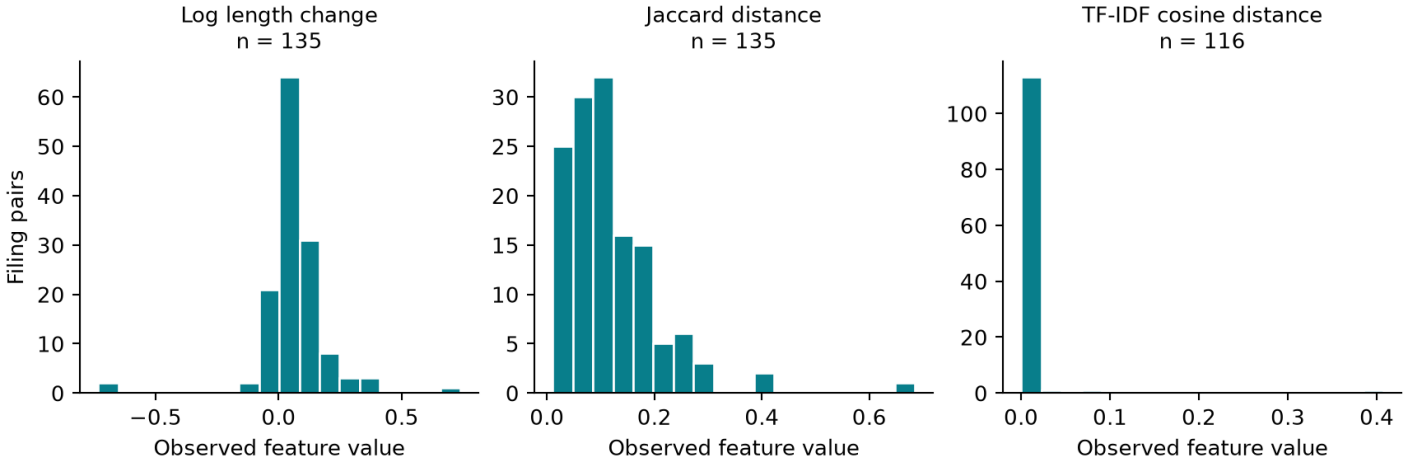
Acceptance year	Annual pairs	Issuers
2015	19	19
2016	19	19
2017	19	19
2018	19	19
2019	19	19
2020	20	20
2021	20	20

The pilot spans 2015-2021; **59** pairs are from 2019-2021. The original plan requested 100-150 pairs in 2019-2021. Coverage over a longer date range does not satisfy that same criterion.

The full study requires a fresh historical population at every year start. The implemented selector uses prior 60-session mean dollar volume, at least 40 valid observations, one most-liquid security per issuer and up to 500 issuers. Later delistings stay in the selected cohort [3].

2. What changed in the text?

Length change measures expansion or contraction in the number of alphabetic tokens. Jaccard distance measures turnover in distinct words. TF-IDF cosine distance weights changes using vocabulary fitted on strictly earlier acceptance years. None is a risk forecast by itself.



Text measure	Observed n	Median	10th / 90th percentile
Current section tokens	135	7601	3448 / 1.268e+04
Log length change	135	0.04595	-0.02304 / 0.1698
Jaccard distance	135	0.09816	0.03452 / 0.2203
TF-IDF cosine distance	116	0.003867	0.00104 / 0.01314

The first 19 pairs have no prior pilot year from which to fit vocabulary, so their TF-IDF values are explicitly unavailable. The remaining 116 distances use earlier years only. This descriptive vocabulary policy does not substitute for the final model's label-maturity filter.

Interpretation limits

Small cosine distances can coexist with meaningful additions to a long section. Token-set distances ignore word order; length does not distinguish a new hazard from added explanation. These summaries justify checking individual passages, not claiming that a model predicts returns.

Example selection is independent of outcomes: the checkpoint selects the smallest SHA-256 of seed and accession within each year, then exposes the source link and a 12-word opening. No largest-return or best-performing example was selected.

3. Timing and audit evidence

For an accepted filing, convert the SEC timestamp to New York time. The reference session is the second XNYS session strictly after its acceptance date. Controls end at the preceding close; the forecast is made before the reference open. Returns begin after the reference close, so the reference day's daily return is excluded [3].

Stage	Evidence and rule
Prior controls	60 exact sessions ending at the preceding close; 20/60 momentum, sample 20-session volatility, log USD capitalization and dated industry
Return label	Product of 1 + daily stock return over the next 20 sessions, minus the same compounded value-weighted market return
Terminal events	Dividends and delisting included once; cash proceeds earn zero after the terminal day while benchmark returns continue
Missing inputs	Unexplained missing sessions, ambiguous security IDs and absent historical snapshots are exclusions or errors; they are never filled with zero

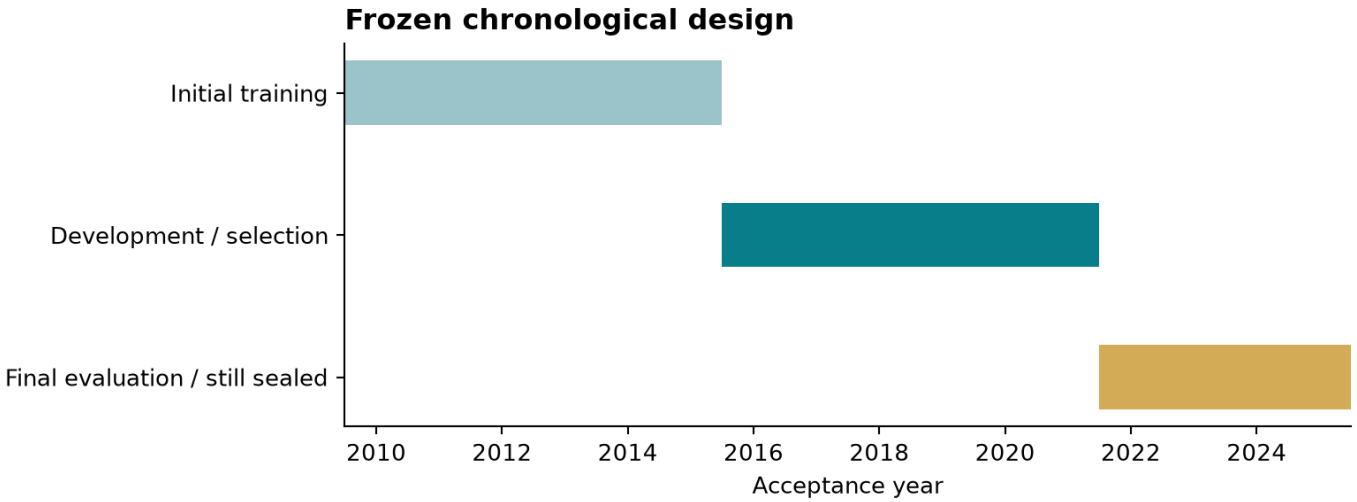
Rebuilt evidence

The source audit records 155 passing filings and 14 registered reference resolutions. It preserves 72 differences between the API-encoded timestamp and original SGML header. The header controls the event date; dissemination time remains an explicit unverified assumption [4].

The matching saved arithmetic receipt records 135 calculated cases and 0 blocked cases, independently reconciled within 1e-12; the maximum absolute difference is 1.43e-15. The fixed 20-trace packet and WFM cash-continuation case remain available. These checks establish arithmetic consistency for candidate mappings, not approval of those mappings [5].

The matched receipts record 0 completed human filing reviews and 0 passing human trace reviews. There are 7 candidate events with a header-CIK difference and 135 links with unresolved open-end markers. These remain M5/M6 completion requirements [5].

4. Model selection and final access



The primary comparison is ridge regression with market controls versus the same regression with all three text-change measures. Each model selects its penalty from 0.1, 1, 10 and 100 by pooled development MSE over six expanding annual folds. This is eight primary configurations, with identical validation observations [3].

The freeze binds development data, predictions, selected penalties, exclusions, source and review evidence, preprocessing code, the Git commit and secondary specifications. Final features and final targets occupy separate files. Final targets are parsed only after the prerequisite and fingerprint checks pass.

For each 2022-2025 acceptance year, the final evaluator refits using earlier events whose label ends strictly precede that year's first prediction. Earlier final-year labels may enter later annual refits automatically under the frozen procedure. Human retuning after final access is prohibited.

Primary score and uncertainty

The score is 1 - SSE(controls + text) / SSE(controls), computed on identical observations. A positive value means lower squared error for the augmented model. The paired bootstrap resamples consecutive three-month reference-month blocks, preserving all firms within a month, with 2,000 draws and a six-month sensitivity.

Four final years contain limited independent time information. The interval is approximate and conditional on the frozen fitted procedure. An undefined zero-baseline-error case is reported explicitly. Forecast accuracy does not establish profitable trading.

5. Robustness and failure analysis

Check	Frozen implementation / interpretation
Year and month	Acceptance-year scores and reference-month Spearman correlations, with missing, small and constant groups reported
Leave one year out	Omit each year from the same frozen predictions; do not retrain using later years
Feature ablations	Six nonempty proper subsets of the three text features, with penalties selected on development data only; length-only is included
Reference delays	Rebuild 1-, 2- and 5-session controls and labels independently from sources; check availability separately; report both own and common event sets
Size and industry	Prior-close covariates and predeclared USD size cutpoints; display small-group limitations
Largest errors	Source-linked descriptive case review; no post-result feature selection
Paragraph changes	Conservative additions, removals, rewording and relocation categories; descriptive until a separate forecast rule is frozen

The 2020 disclosure-rule transition

The SEC's 2020 amendments added a summary requirement for long risk-factor sections and relevant-heading requirements [2]. Changes in structure can change text metrics without equivalent changes in economic risk. The declared sensitivity inspects transition-period examples on development data and reports diagnostics without choosing a new primary specification.

No empirical robustness conclusion exists yet. The software reports negative, undefined and unavailable checks as such. Empty sections in the final results are not evidence of a null finding.

6. Reproduction and build verification

The public package contains the locked Python environment, tests, an invented full-study demonstration, source-derived aggregate pilot exhibits, and explicit evidence contracts. Licensed market records and their event-level return values remain private.

Evidence	Recorded status
Original test baseline	196 passing tests before this implementation
Current test run	361
Recovered SEC audit	155 / 155 automated checks; no blocking source differences
Recovered candidate arithmetic	135 / 135 independently reconciled; no identity approval inferred
End-to-end demonstration	Passed: 192 invented memberships; 144 development and 48 final observations; all four robustness components completed
Empirical final outcomes	Not opened; no final score or confidence interval estimated

Reproduction commands

`make reproduce-full`
Runs the frozen-protocol checks, integrity tests and the synthetic historical-universe-to-final-evaluation demonstration. Every generated fixture is marked synthetic.

`make pilot-checkpoint`
Rebuilds the SEC-only descriptive pilot from the private source checkpoint and a matching source audit.

`make research-report`
Rebuilds the figures and this progress report from saved public aggregate pilot evidence and the verification receipt.

Code and drafting were co-developed with Codex. Automated and assistant checks remain distinct from a person's review and an independent reproduction. A reviewer should challenge timestamp assumptions, exclusion mechanisms, and the interpretation of every reported quantity.

7. Completion criteria and sources

Milestone	Remaining empirical completion requirement
6 - Pilot checkpoint	Reviewed source/identity evidence, controls-only smoke on approved 2019-2021 rows, and another person's reproduction
7 - Full dataset	Full historical listing/liquidity/link/return exports and filing ingestion; second 30-pair audit; explained coverage by year
8 - Model freeze	Run development on the approved full dataset; freeze exact evidence, penalties, code and secondary specification
9 - Final evaluation	Execute the frozen 2022-2025 evaluation once after all access gates pass
10 - Robustness	Execute declared checks on the frozen empirical predictions and audit failure cases
11 - Public research package	Replace pending results with actual estimates, complete independent reproduction, and publish the final interpretation

Semantic secondary comparator

A separate MiniLM checkpoint is pinned by revision and weight hash. Its observed revision is dated 2026, so the comparator is explicitly retrospective. Verified pretraining and fine-tuning cutoffs are unavailable. The encoder processes every token in non-overlapping chunks and records zero truncation; manual unchanged/reworded/new-risk checks remain pending [6].

Sources and audit trail

[1] Cohen, Malloy and Nguyen (2020), *Lazy Prices*, Journal of Finance 75(3). doi.org/10.1111/jofi.12885
[2] SEC (26 August 2020), Regulation S-K amendments. sec.gov/newsroom/press-releases/2020-192
[3] Frozen protocol v1.0.0 and configs/protocol.json.
[4] Milestone 04 source audit; pilot_checkpoint/pilot_checkpoint.json.
[5] Pilot arithmetic receipt; review_followup_summary.json.
[6] Pinned MiniLM checkpoint; configs/semantic_retrospective.json.